

Modelling the informal sector formally

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Abstract: The theoretical characterization of formal–informal sector dualism by labor-market dualism is integrated with its empirical characterization by size dualism by supposing that the minimum formal sector wage is only enforced for firms greater than a certain size. A firm's size varies directly with the talent of its entrepreneur–manager, and the choice between formal and informal sector entrepreneurship is determined endogenously. It is then shown that the size gap between the smallest formal sector firm and largest informal sector firm varies directly with the formal–informal sector wage differential, which in turn is shown as expected to increase the further is the minimum wage above the market-clearing wage. The comparative statics of changes in the size of the firm above which the minimum wage is enforced are also examined.

1. Introduction

The part of the urban economy in less developed countries that was later to become known as 'the informal sector' was not initially seen by economists as worthy of separate study. For Lewis (1954) it was merely one of the sources of an infinitely elastic supply of labor to the 'capitalist sector' at the prevailing real wage. For Todaro (1969) it was a way station on route to a job in the urban modern sector, again at the prevailing real wage. Thus Todaro (p. 139) 'views migration as a two-stage phenomenon. The first stage finds the unskilled rural worker migrating to an urban area and initially spending a certain period of time in the so-called "urban traditional" sector. The second stage is reached with the eventual attainment of a more permanent modern sector job'. The prevailing wage is not bid down either because it is already at a subsistence floor (Lewis), or because it is politically determined [Harris and Todaro (1970)], or because of 'efficiency wage' considerations [surveyed by Stiglitz (1982)].

This view of the informal sector as a holding ground for workers awaiting entry into the formal sector was challenged by the International Labour Office mission to Kenya (1973). Its report instead saw the informal sector as a source of employment in its own right and as a category for economic planning. The ILO went on to commission a series of studies of Third World cities, which were summarized in a book edited by Sethuraman (1981). In

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most of these studies the informal sector was in practice defined as consisting of private enterprises with ten or fewer employees. The book concluded that 'there is little evidence to show that the participants in this sector hold out for better jobs in the formal sector. In other words, virtually all of them seem to view the informal sector as a permanent source of employment and income' (p. 198). Moreover, a great deal of 'reverse' mobility was observed in the form of formal sector employees becoming informal sector entrepreneurs, which was consistent with the evidence that 'a majority of the entrepreneurs earned an income substantially higher than that implied by the legal minimum wage' (p. 195).

This evidence gave rise to a view opposite from the economists' initial one. The new view was that the informal sector is simply the small-scale end of an urban economic continuum. Thus a recent writer advocates that the concept of the informal sector be jettisoned altogether [Peattie (1987)]. Yet the basic labor market dualism from the original economists' conception is still supported by the evidence. The above study concludes about 'the informal sector as a whole' that 'a majority of the wage-earning employees ... receive wages below the legal minimum' (p. 193) which is enforced in the formal sector. In his World Bank study, Mazumdar (1976) agrees that, 'the basic distinction between the two sectors turns on the idea that employment in the formal sector is in some sense or senses protected so that the wage-level and working conditions in the sector are not available, in general, to the job-seekers in the market unless they manage to cross the barrier of entry somehow. This kind of "protection" may arise from the action of trade unions, of governments, or of both acting together' (p. 656). In a more recent study Banerjee (1983) finds, for workers in Delhi, 'when an earnings function is estimated for all wage employees with a dummy variable indicating employment in the formal sector included among the explanatory variables, that coefficient is positive and significant at the one percent level' (p. 450).

In the empirical literature cited above the informal sector is always defined as consisting of economic units less than or equal to a certain size measured by number of employees. For example in the Banerjee study, 'employees in government and public sector establishments, and in privately owned establishments employing 20 or more workers are assigned to the formal sector. All other workers belong to the informal sector'. Let us define as *size dualism* the existence of a difference in size between the smallest formal sector firm and the largest informal sector firm. In contrast, the theoretical literature focuses on *labor market dualism*, which we define as the existence of a difference between the formal sector wage and the informal sector wage for economically identical employees. We can integrate the empirical and theoretical characterizations of formal-informal sector dualism if we suppose that the aforementioned 'prevailing real wage' is enforced only for firms that are larger than a certain size measured by number of employees. This

qualification would be consistent with many of the hypotheses advanced to explain the formal sector wage floor. If it is the government's legal minimum wage that is binding, it is clear, given the limited administrative resources of Third World governments, that they can maximize the number of workers covered by the law by focusing enforcement efforts on large firms. If it is union bargaining power that sets the wage floor, it is well known that unions concentrate their organizing efforts on large firms. This can be explained by economies of scale in organizing due to fixed costs. Even if one believes the efficiency wage hypothesis is the relevant one, if the efficiency wage results from minimizing the costs of monitoring workers' effort [Shapiro and Stiglitz (1984)], it is reasonable to think that small firms can monitor nearly costlessly and therefore do not need to pay the efficiency wage.¹

In the next section we show in a model where labor is the only factor of production that the assumption of a minimum wage enforced only for firms greater than a certain size is sufficient to give rise to formal-informal sector dualism, provided that the minimum wage is binding. In particular, there is a break in the size distribution of firms (rather than a continuum) between those which pay the minimum wage and those which do not and there is a market determined wage for the latter firms which is lower than the minimum wage. The model is consistent with some (but not all) informal sector entrepreneurs earning more than the minimum wage. In section 3 we examine the comparative statics of changes in the minimum wage and changes in the size of firm above which the minimum wage is enforced. Section 4 presents some brief conclusions.

2. A model of formal-informal sector dualism

To begin with, we need a model that in the absence of market distortions generates an 'urban economic continuum' of firm sizes. Here we turn to a paper of Lucas (1978). In this paper the size distribution of firms reflects the underlying distribution of managerial (or entrepreneurial) talent, with greater talent generating a larger firm size by Hicks-neutrally shifting a decreasing returns to scale production function. Obviously not every individual can be a manager, so a cutoff level of managerial talent emerges at the level where managerial rent just equals the wage (the opportunity cost of the manager's time).

Formally, assume that every agent in the economy is endowed with a managerial (or entrepreneurial) talent level x drawn from a fixed distribution $D: R^+ \rightarrow [0, 1]$. Assume further that there is a continuum of agents so that the entire distribution D of talent is always fully represented. A firm consists of a

¹A recent empirical study by Brown and Medoff (1989) casts doubt on the monitoring cost explanation of the positive correlation between wages and firm size in the United States.

single manager (entrepreneur) of talent x and $N(x)$ homogeneous employees. Each firm's output of the composite commodity (which serves as the numeraire) is given by

$$Q(x) = xG[N(x)], \quad (1)$$

where G is twice-differentiable, increasing, strictly concave and satisfies $G(0) = 0$, x is the managerial talent level of a type x agent, and $Q(x)$ is the level of output of the firm being managed by a type x agent. G is assumed to display diminishing returns due to the manager's limited 'span of control'. The income (rent) to a type x manager $r(x)$ equals $xG[N(x)] - wN(x)$, where w equals the wage rate that under perfect competition is the same for all employees since their talent is relevant only if they manage. The cutoff level of managerial talent z is determined when this rent is just equal to the opportunity cost of the manager's time w :

$$r(z) \equiv zG[N(z)] - wN(z) = w. \quad (2)$$

We want to show that a unique market wage is determined in this economy and that a universally enforced minimum wage set above the level of this market wage will cause unemployment. The first-order condition for rent (profit) maximization yields

$$xG'[N(x)] = w, \quad x \geq z. \quad (3)$$

Full employment of the population N requires that the fraction who manage plus the fraction who are employees must sum to one:

$$1 - D(z) + \int_z^\infty N(x) dD(x) = 1.$$

We can rewrite this full employment condition to state that demand for labor must equal the supply:

$$\int_z^\infty N(x) dD(x) = D(z). \quad (4)$$

Solving eq. (3) for $N(x)$, we have

$$N(x) = G'^{-1}(w/x) \equiv H(w/x), \quad (5)$$

where since G is strictly concave it can be shown that H is monotonically

decreasing in its argument. Intuitively, it is now clear that eq. (4) determines a unique competitive equilibrium w . On the right-hand side, labor supply increases with w as more individuals choose to become employees rather than managers, while on the left-hand side, labor demand decreases with w as the number of managers declines and each remaining manager hires fewer workers. The labor supply and demand curves must intersect since everyone chooses to become a worker as $w \rightarrow \infty$ and to become a manager as $w \rightarrow 0$. This proof is formalized in the appendix.

Now suppose that a minimum wage $\bar{w}$ that exceeds the competitive equilibrium wage is universally enforced. This will unambiguously increase the number of people wanting to be employees [the right-hand side of eq. (4)] and decrease the demand for labor [the left-hand side of eq. (4)], leading to an excess supply of job-seekers. If, however, the minimum wage is not enforced for firms less than or equal to a certain size $\bar{N}$ (twenty employees or fewer, say), then firms less than or equal to this size offering a wage less than $\bar{w}$ will form. These firms will constitute the informal sector, while the other firms make up the formal sector. Note that in this new equilibrium no one is *unemployed*, although the workers in the informal sector can be said to be *underemployed* in the sense that they would be earning more in competitive equilibrium.

Given the conditions of the preceding paragraph it is obvious that the informal sector wage, which we denote by w^* , must be less than $\bar{w}$. In the introduction, however, we cited evidence that many (but not all) informal sector *entrepreneurs* earn *more* than $\bar{w}$. In this case the relevant opportunity cost for formal sector managers is not the formal sector wage but rather the rent they could earn in the informal sector. Intuitively, it is clear that the talent level z of the marginal formal sector manager will then be determined by the point at which the advantage of being able to hire the profit-maximizing amount of labor just balances the disadvantage of having to pay the higher formal sector wage. Mathematically, z is determined by the condition

$$r(z) \equiv zG[N(z)] - \bar{w}N(z) = zG(\bar{N}) - w^*\bar{N} \equiv r^*(z). \quad (6)$$

Fig. 1 shows that a z satisfying eq. (6) exists and is unique. To construct this figure, note that $r(x)$ is precisely analogous to the standard profit function of microeconomic theory, with the managerial talent level x playing the role of the price p faced by a perfectly competitive firm. We know that the derivative of the profit function with respect to p gives us output supply, so it follows that the derivative of $r(x)$ with respect to x gives us $G[N(x)]$. (Alternatively, this result can be derived directly using the envelope theorem.) Since from eq. (5) we know that $N(x)$ is increasing in x given $\bar{w}$, it follows that the slope of $r(x)$ in fig. 1 is always increasing. For $0 \leq x \leq \bar{x}$, where $\bar{x}$ is

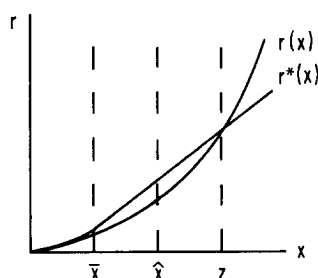


Fig. 1

determined by $H(w^*/\bar{x}) = \bar{N}$, $r^*(x)$ is just the same 'profit function' as $r(x)$ only evaluated at the lower wage w^* . Therefore, as shown in fig. 1, $r^*(x) > r(x)$ for $0 < x \leq \bar{x}$ and the slope of $r^*(x)$ is steeper than the slope of $r(x)$ over this interval. For $x > \bar{x}$, however, the slope of $r^*(x)$ is fixed at $G(\bar{N})$. The slope of $r(x)$ catches up to the slope of $r^*(x)$ for $\hat{x} > \bar{x}$, where $\hat{x}$ is determined by $H(\bar{w}/\hat{x}) = \bar{N}$. Clearly $r^*(\hat{x}) > r(\hat{x})$ since output is the same but informal sector wage payments are less. For $x > \hat{x}$, however, $r(x)$ increases at a faster rate than $r^*(x)$, so as fig. 1 illustrates there must exist one and only one talent level z satisfying $r(z) = r^*(z)$.

We must have $z > \hat{x}$, from which it follows that $N(z) > \bar{N}$: labor market dualism begets size dualism. Moreover, we can show that this difference between the size of the smallest formal sector firm and the largest informal sector firm increases as the formal-informal sector wage gap increases. Since $\bar{w}$ and $\bar{N}$ are constant, we need only show that as w^* decreases, z increases, thus increasing $N(z)$ and $N(z) - \bar{N}$. But this result follows immediately from fig. 1: a decrease in w^* shifts up the $r^*(x)$ schedule, causing it to intersect the $r(x)$ schedule at a higher z .² The intuitive explanation for this result is simple. As w^* falls, the disadvantage of having to pay the higher formal sector wage rises. A higher level of managerial talent is therefore needed to generate a sufficient advantage from being able to hire the profit-maximizing amount of labor to overcome this increased disadvantage of operating in the formal sector. Thus the talent level of the marginal formal sector manager rises, and so does the level of employment of the smallest formal sector firm, given that the formal sector wage is fixed. In sum, *the greater the extent of labor market dualism, the greater is the extent of size dualism.*

3. Comparative statics of formal sector policy

It would be interesting to know what are the effects of changes in the

²Note that as w^* increases towards $\bar{w}$, $\bar{x}$ and z both converge towards $\hat{x}$.

formal sector wage $\bar{w}$ or the maximum size of informal sector firm $\bar{N}$ on the informal sector wage w^* . Intuitively, one would expect that w^* would vary inversely with $\bar{w}$, since an increase in $\bar{w}$, say, would cause a contraction in formal sector employment and a fall in w^* as the displaced workers entered the informal sector. Similarly, one would expect w^* to vary directly with $\bar{N}$, since an increase in $\bar{N}$, say, would cause an expansion of demand for informal sector labor, driving up w^* . However, these clear effects working through demand for hired labor are complicated by ambiguous effects resulting from reallocation of managers between the formal and informal sectors. Further analysis is therefore required to know when the clear effects dominate the ambiguous ones.

To do precise comparative static analysis we first need to show that a unique w^* exists. We can use the same procedure that we used above to show the existence of a unique free-market equilibrium wage. In free-market equilibrium we divided the population into only two classes: managers (entrepreneurs) and employees. Now it must be divided into formal sector managers, formal sector employees, informal sector managers, and informal sector employees. Moreover, it is necessary to distinguish between informal sector managers earning more than $\bar{w}$ and their employees and informal sector managers earning less than $\bar{w}$ and their employees. The reason is that the latter group of informal sector managers would prefer to be formal sector employees if such employment were available.

Specifically, let $\bar{z}$ be determined by

$$\bar{z}G[N(\bar{z})] - w^*N(\bar{z}) = \bar{w} \quad (7)$$

(implicitly assuming $N(\bar{z}) < \bar{N}$) and let z^* be determined by

$$z^*G[N(z^*)] - w^*N(z^*) = w^*. \quad (8)$$

Everyone with managerial talent $\geq \bar{z}$ becomes a manager in either the formal or informal sector, while everyone with managerial talent $< z^*$ becomes an employee in either the formal or informal sector. However, everyone with managerial talent $< \bar{z}$ desires formal sector employment. The proportion of those desiring formal sector employment who actually receive it is $\int_{\bar{z}}^{\infty} N(x) dD(x)/D(\bar{z})$. Suppose these formal sector employees are chosen at random from the population with managerial talent $< \bar{z}$. Then the fraction of the population with talent between $\bar{z}$ and z^* who obtain formal sector employment is $[\int_{\bar{z}}^{\infty} N(x) dD(x)/D(\bar{z})][D(\bar{z}) - D(z^*)]$ and the fraction who become informal sector entrepreneurs earning less than $\bar{w}$ is $[1 - \int_{\bar{z}}^{\infty} N(x) dD(x)/D(\bar{z})][D(\bar{z}) - D(z^*)]$.

Summing up the fractions of the population accounted for, respectively, by formal sector managers (A), their employees (B), informal sector managers

earning more than $\bar{w}$ (C), their employees (D), informal sector managers earning less than $\bar{w}$ (E), and their employees (F), we have

$$\begin{aligned}
 & \text{A} \qquad \text{B} \qquad \text{C} \qquad \text{D} \\
 & [1 - D(z)] + \int_z^\infty N(x) dD(x) + [D(z) - D(\bar{z})] + \left\{ [D(z) - D(\bar{x})] \bar{N} + \int_{\bar{z}}^{\bar{x}} N(x) dD(x) \right\} \\
 & \qquad \qquad \qquad \text{E} \\
 & \qquad \qquad \qquad + [D(\bar{z}) - D(z^*)] \left[1 - \int_z^\infty N(x) dD(x) / D(\bar{z}) \right] \\
 & \qquad \qquad \qquad \text{F} \\
 & \qquad \qquad \qquad + \int_{z^*}^{\bar{z}} N(x) dD(x) \left[1 - \int_z^\infty N(x) dD(x) / D(\bar{z}) \right] = 1.
 \end{aligned}$$

A + B equals the size of the formal sector and C + D + E + F equals the size of the informal sector, so by keeping track of what happens to A + B we can see how the size of the formal relative to the informal sector changes when $\bar{w}$ and $\bar{N}$ change in addition to seeing how w^* changes. We can simplify and rearrange this full employment condition to get the equivalent of eq. (4):

$$\begin{aligned}
 & \text{B} \qquad \text{D} \\
 & \int_z^\infty N(x) dD(x) + [D(z) - D(\bar{x})] \bar{N} + \int_{\bar{z}}^{\bar{x}} N(x) dD(x) \\
 & \qquad \qquad \qquad \text{E + F} \\
 & + \left[D(\bar{z}) - D(z^*) + \int_{z^*}^{\bar{z}} N(x) dD(x) \right] \left[1 - \int_z^\infty N(x) dD(x) / D(\bar{z}) \right] = D(\bar{z}). \tag{9}
 \end{aligned}$$

Eq. (9) states that the fraction of the population desiring formal sector employment equals the sum of the fraction that obtains formal sector employment (B) and the fractions that must settle for informal sector employment (D and F) or informal sector entrepreneurship earning less than $\bar{w}$ (E).

In order to see whether eq. (9) determines a unique w^* , we first note from eqs. (7) and (8), respectively, that $\bar{z}$ and z^* vary directly with w^* , while we already know that z varies inversely with w^* . As w^* approaches $\bar{w}$ from below, we can see from eqs. (6)–(8) that z^* approaches $\bar{z}$ from below while z and $\bar{z}$ converge, so that at $w^* = \bar{w}$ all terms on the left-hand side of (9) except

B (formal sector employment) vanish. But we previously assumed that at $\bar{w}$ there is an excess supply of individuals seeking employment, so the right-hand side of (9) must exceed the left-hand side at $\bar{w}$. To insure existence of a w^* satisfying (9), it is now also assumed that there is excess demand for labor at $w^* = 0$. Does the excess supply existing when $w^* = \bar{w}$ decrease monotonically as w^* falls below $\bar{w}$? The supply of individuals seeking employment falls ($\bar{z}$ decreases) since informal sector entrepreneurship becomes more attractive. On the demand side, each informal sector entrepreneur hiring less than $\bar{N}$ workers expands her employment, increasing terms D and F. If it were not for the effect of reallocating managers, then, the fall in w^* would unambiguously decrease the excess supply of labor. In particular, as w^* falls z increases, decreasing B unambiguously, while increasing C by shifting managers to the informal sector where they each employ $\bar{N}$ workers, and also increasing E and F by decreasing the chance of obtaining a formal sector job, leading to an increase in informal sector entrepreneurs earning less than $\bar{w}$. The total effect of the rise in z on the excess supply of labor is shown in the appendix to be ambiguous: the effect tends to be negative, the smaller the difference between the size of the smallest formal sector firm $N(z)$ and $\bar{N}$ and the smaller the proportion of informal sector participation accounted for by employment by managers earning more than $\bar{w}$. On the other hand, the appendix also shows that the effects of decreasing $\bar{z}$ and z^* on the excess supply of labor are unambiguously negative. If these latter two effects do not more than offset the effect of increasing z (if it needs to be offset), then sufficient responsiveness of demand for labor $H(w^*/x)$ to the fall in w^* surely will.

We can now carry out our comparative static analysis of the effects of changes in $\bar{w}$ and $\bar{N}$. In particular, if an increase in $\bar{w}$ positively affects excess supply of labor then a decrease in w^* will be necessary to maintain the equality in (9), and if an increase in $\bar{N}$ negatively affects excess supply of labor then an increase in w^* will be necessary to maintain this equality. Looking at eqs. (6)–(8), we can see that z and $\bar{z}$ increase with $\bar{w}$ while z^* is unaffected. Turning to eq. (9), the increase in $\bar{z}$ means the supply of individuals seeking jobs increases, reflecting the increased attractiveness of formal sector employment relative to formal or informal sector entrepreneurship. At the same time each formal sector entrepreneur reduces her hiring, decreasing B. Once again, ambiguity is introduced only by the induced reallocation of managers. It is shown in the appendix that sufficient responsiveness of demand for labor $H(\bar{w}/x)$ insures that the overall effect of the increase in $\bar{w}$ on excess supply of labor is positive, so that $dw^*/d\bar{w} < 0$. Eqs. (6)–(8) also tell us that z increases with $\bar{N}$, since the attractiveness of informal relative to formal sector entrepreneurship increases, while $\bar{z}$ and z^* are unchanged, because no one on the margin between being a manager and being an employee is affected. Turning again to eq. (9), the supply of

individuals seeking jobs is unchanged but the increase in $\bar{N}$ increases informal sector employment D by allowing those informal sector entrepreneurs who desired to hire more workers to do so. This negative effect on the excess supply of labor may be reinforced or offset by the managerial reallocation effect resulting from the increase in z . Even in the latter case it may still be that the total effect of the increase in $\bar{N}$ on the excess supply of labor is negative so that $dw^*/d\bar{N} > 0$, but this result cannot be insured by sufficient responsiveness of labor demand to wage changes as can $dw^*/d\bar{w} < 0$.

We can sum up our comparative static results as follows:

- (i) Sufficient responsiveness of labor demand will insure (though it is probably not necessary) that the informal sector wage will vary inversely with the formal sector wage. Thus the formal-informal sector wage differential increases with the formal sector wage, and it follows that the extent of size dualism does as well. Moreover, the size of the formal sector varies inversely (and hence the size of the informal sector varies directly) with the formal sector wage: this follows since z varies directly and $H(\bar{w}/x)$ varies inversely with $\bar{w}$, causing the sum of the terms A and B in eq. (9) to vary inversely with $\bar{w}$.
- (ii) If the effect of $\bar{N}$ on informal sector employment is reinforced by or dominates its managerial reallocation effect, the informal sector wage will vary inversely with enforcement of the formal sector wage, i.e. if the government and/or labor unions allow the size of the firm above which the minimum wage is enforced to increase then the informal sector wage will increase.

4. Conclusions

In the process of modelling the informal sector formally we have managed to cover a good deal of microeconomic detail without losing analytical tractability. We are able to analyze the effects of changes in the parameters of the model on the incomes of four different groups of people: formal sector entrepreneurs, formal sector employees, informal sector entrepreneurs, and informal sector employees. Within the group of informal sector entrepreneurs we can distinguish between those who are and those who are not better off than formal sector employees.

A promising candidate for future research would be integration of the model presented in this paper with a model of rural-urban migration. Migration for an individual with managerial talent below the cutoff level z^* would presumably then take place in response to a probability-weighted average of the formal and informal sector wage rates, rather than in response to a probability-weighted formal sector wage rate alone as in the aforemen-

tioned paper of Harris and Todaro.³ This may have implications for the effects of, for example, a formal sector wage subsidy on migration, since such a subsidy would cause a rise in the informal sector wage rate and thus exert an even stronger pull on rural migrants than the Harris–Todaro model predicts. It would also be interesting to see what effects the formal–informal sector dichotomy has in a dynamic context where we allow the possibility of ‘inheritance’ of formal and informal sector status, for example due to ‘connections’ made by formal sector parents. This inheritance could be a powerful force tending to generate dynastic inequality. It is hoped that many such refinements in our understanding of the working of LDC economies will be facilitated by incorporation of the informal sector into formal analysis.

Appendix

Determining a unique competitive equilibrium w

Begin with eq. (4) in the text. We can find z as a monotonically increasing function of w using eq. (2): note that the left-hand side of (2) is increasing in z (use the envelope theorem) so that as w increases, z must increase. Moreover, we have $z=0$ when $w=0$ and $z \rightarrow \infty$ as $w \rightarrow \infty$. Substituting this result and eq. (5) into eq. (4) gives us

$$\int_{z(w)}^{\infty} H(w/x) dD(x) = D[z(w)].$$

The right-hand side of this equation increases monotonically from 0 towards 1 as w increases from 0 towards ∞ , while the left-hand side tends to 0 as w tends to ∞ and increases monotonically as w decreases to 0. Clearly then this equation determines a unique, positive w .

*Determining a unique informal sector wage w^**

Begin with eq. (9) in the text. Substituting in eq. (5) gives us

$$\int_z^{\infty} H(\bar{w}/x) dD(x) + [D(z) - D(\bar{x})]\bar{N} + \int_{\bar{z}}^{\bar{x}} H(w^*/x) dD(x)$$

³Models of rural–urban migration incorporating formal and informal urban sectors in which all individuals are employees have been developed by Fields (1975, 1989).

$$+ \left[D(\bar{z}) - D(z^*) + \int_{z^*}^{\bar{z}} H(w^*/x) dD(x) \right] \left[1 - \int_z^{\infty} H(\bar{w}/x) dD(x)/D(\bar{z}) \right] - D(\bar{z}) = 0. \quad (\text{A.1})$$

The left-hand side of eq. (A.1) gives excess demand for labor, which we want to decrease monotonically as w^* increases. Totally differentiating the left-hand side with respect to w^* gives us

$$\begin{aligned} & -H(\bar{w}/z) dD(z) dz/dw^* + dD(z) \bar{N} dz/dw^* - dD(\bar{x}) \bar{N} d\bar{x}/dw^* \\ & + H(w^*/\bar{x}) dD(\bar{x}) d\bar{x}/dw^* \\ & - H(w^*/\bar{z}) dD(\bar{z}) d\bar{z}/dw^* + \int_{z^*}^{\bar{z}} H'(w^*/x)(1/x) dD(x) \\ & + \left[dD(\bar{z}) d\bar{z}/dw^* - dD(z^*) dz^*/dw^* + H(w^*/\bar{z}) dD(\bar{z}) d\bar{z}/dw^* \right. \\ & \left. - H(w^*/z^*) dD(z^*) dz^*/dw^* + \int_{z^*}^{\bar{z}} H'(w^*/x)(1/x) dD(x) \right] \\ & \times \left[1 - \int_z^{\infty} H(\bar{w}/x) dD(x)/D(\bar{z}) \right] \\ & + \left[D(\bar{z}) - D(z^*) + \int_{z^*}^{\bar{z}} H(w^*/x) dD(x) \right] \left[D(\bar{z}) H(\bar{w}/z) dD(z) dz/dw^* \right. \\ & \left. + \int_z^{\infty} H(\bar{w}/x) dD(x) \cdot dD(\bar{z}) d\bar{z}/dw^* \right] / [D(\bar{z})]^2 - dD(\bar{z}) d\bar{z}/dw^*. \end{aligned}$$

Since $\bar{N} = H(w^*/\bar{x})$ the third and fourth terms cancel. We now simplify by grouping the terms into those reflecting the direct effect of the increase in w^* on the hiring by individual informal sector managers [terms including

$H'(w^*/x)]$ and those reflecting reallocation of managers (terms including dz/dw^* , $d\bar{z}/dw^*$, or dz^*/dw^*). We have

$$\begin{aligned}
 & \int_{\bar{z}}^{\bar{x}} H'(w^*/x)(1/x) dD(x) + \int_{z^*}^{\bar{z}} H'(w^*/x)(1/x) dD(x) \left[1 - \int_z^{\infty} H(\bar{w}/x) dD(x)/D(\bar{z}) \right] \\
 & + \left[-H(\bar{w}/z) \left\{ 1 - \left[D(\bar{z}) - D(z^*) + \int_{z^*}^{\bar{z}} H(w^*/x) dD(x) \right] / D(\bar{z}) \right\} + \bar{N} \right] dD(z) dz/dw^* \\
 & - \left[\int_z^{\infty} H(\bar{w}/x) dD(x)/D(\bar{z}) \right] \\
 & \times \left\{ 1 - \left[D(\bar{z}) - D(z^*) + \int_{z^*}^{\bar{z}} H(w^*/x) dD(x) \right] / D(\bar{z}) + H(w^*/\bar{z}) \right\} dD(\bar{z}) d\bar{z}/dw^* \\
 & - [1 + H(w^*/z^*)] \left[1 - \int_z^{\infty} H(\bar{w}/x) dD(x)/D(\bar{z}) \right] dD(z^*) dz^*/dw^*. \tag{A.2}
 \end{aligned}$$

Consider the expression $1 - [D(\bar{z}) - D(z^*) + \int_{z^*}^{\bar{z}} H(w^*/x) dD(x)]/D(\bar{z})$ which appears twice in eq. (A.2). Using eq. (A.1), we have

$$\begin{aligned}
 & \left[D(\bar{z}) - D(z^*) + \int_{z^*}^{\bar{z}} H(w^*/x) dD(x) \right] / D(\bar{z}) \\
 & = D(\bar{z}) - \int_z^{\infty} H(\bar{w}/x) dD(x) - [D(z) - D(\bar{x})] \bar{N} - \int_{\bar{z}}^{\bar{x}} H(w^*/x) dD(x) \\
 & \quad \div D(\bar{z}) \left[1 - \int_z^{\infty} H(\bar{w}/x) dD(x)/D(\bar{z}) \right] \\
 & = 1 - \left\{ [D(z) - D(\bar{x})] \bar{N} + \int_{\bar{z}}^{\bar{x}} H(w^*/x) dD(x) \right\} \\
 & \quad / \left[D(\bar{z}) - \int_z^{\infty} H(\bar{w}/x) dD(x) \right],
 \end{aligned}$$

so that

$$1 - \left[D(\bar{z}) - D(z^*) + \int_{z^*}^{\bar{z}} H(w^*/x) dD(x) \right] / D(\bar{z})$$

$$= \left\{ [D(z) - D(\bar{x})] \bar{N} + \int_{\bar{z}}^{\bar{x}} H(w^*/x) dD(x) \right\} / \left[D(\bar{z}) - \int_{\bar{z}}^{\infty} H(\bar{w}/x) dD(x) \right] > 0.$$

We can now see that the term in eq. (A.2) involving reallocation of managers between the formal and informal sectors (the term including dz/dw^*) is the only one that is not unambiguously negative. This term would vanish if, for example, the size of the smallest formal sector firm were twice that of the largest informal sector firm and employment by informal sector managers earning more than $\bar{w}$ accounted for half of total informal sector participation (not counting those managers themselves). If this term is positive and not offset by the next two terms, we can still insure that the sum of all five terms is negative by choosing our production function to make $H'(w^*/x)$, and hence the first two terms, large enough.

*Comparative static effect of $\bar{w}$ on w^**

Totally differentiating eq. (A.1) with respect to $\bar{w}$ gives us the third and fourth terms from eq. (A.2), replacing dz/dw^* , $d\bar{z}/dw^*$ with $dz/d\bar{w}$, $d\bar{z}/d\bar{w}$ while the other terms are replaced by

$$\int_{\bar{z}}^{\infty} H'(\bar{w}/x)(1/x) dD(x) \left\{ 1 - \left[D(\bar{z}) - D(z^*) + \int_{z^*}^{\bar{z}} H(w^*/x) dD(x) \right] / D(\bar{z}) \right\}.$$

This new term and the term including $d\bar{z}/d\bar{w}$ are unambiguously negative while the sign of the term including $dz/d\bar{w}$ is ambiguous but in the opposite direction from that in eq. (A.2) since $dz/d\bar{w} > 0$ while $dz/dw^* < 0$. We can insure that the sum of the three terms is negative in the same way as we did for eq. (A.2).

*Comparative static effect of $\bar{N}$ on w^**

Totally differentiating eq. (A.1) with respect to $\bar{N}$ gives us once again the term in eq. (A.2) involving reallocation of managers between the formal and informal sectors, this time replacing dz/dw^* with $dz/d\bar{N}$, while the remaining terms are replaced by $D(z) - D(\bar{x})$, which is clearly positive. This positive term may be reinforced or offset by the ambiguously-signed reallocation term. In the latter case, there is no way to insure that the sum of the two terms will be positive.

References

- Banerjee, B., 1983, The role of the informal sector in the migration process: A test of probabilistic migration models and labour market segmentation for India, *Oxford Economic Papers* 35, 399–422.
- Brown, C. and J. Medoff, 1989, The employer size wage effect, *Journal of Political Economy*, forthcoming.
- Fields, G.S., 1975, Rural–urban migration, urban unemployment and under-employment, and job-search activity in LDCs, *Journal of Development Economics* 2, 165–187.
- Fields, G.S., 1989, On-the-job search in a labor market model: Ex ante choices and ex post outcomes, *Journal of Development Economics* 30, 159–178.
- Harris, J.R. and M.P. Todaro, 1970, Migration, unemployment, and development: A two-sector analysis, *American Economic Review* 60, 126–142.
- International Labour Office, 1973, *Employment, incomes and equality: A strategy for increasing productive employment in Kenya* (International Labour Office, Geneva).
- Lewis, W.A., 1954, Economic development with unlimited supplies of labor, *The Manchester School* 22, 139–191.
- Lucas, R.E., 1978, On the size distribution of business firms, *Bell Journal of Economics* 9, 508–523.
- Mazumdar, D., 1976, The urban informal sector, *World Development* 4, 655–679.
- Peattie, L., 1987, An idea in good currency and how it grew: The informal sector, *World Development* 15, 851–860.
- Sethuraman, S.V., ed., 1981, *The urban informal sector in developing countries* (International Labour Office, Geneva).
- Shapiro, C. and J.E. Stiglitz, 1984, Equilibrium unemployment as a worker discipline device, *American Economic Review* 74, 433–444.
- Stiglitz, J.E., 1982, Alternative theories of wage determination and unemployment: The efficiency wage model, in: M. Gersovitz et al., eds., *The theory and experience of economic development* (Allen and Unwin, London) 78–106.
- Todaro, M.P., 1969, A model of labor migration and urban unemployment in less developed countries, *American Economic Review* 59, 138–148.